DISGT: Dynamic-Interactive Subgraph Transformer for Unsupervised Hyperspectral Band Selection

Chong Cheng, Mingwei Wang, Kaixiong Wu , Wei Liu, Zeyu Tang, and Maolin Chen

***Abstract*— Band selection (BS) is one of the effective ways to reduce data dimensionality while retaining important information in hyperspectral imagery (HSI). The ultimate goal is to represent the original HSI using as few informative and discriminative bands as possible. However, existing BS methods usually divide all bands into independent spaces while neglecting the relationships between these spaces. Moreover, the evaluation of band importance through its neighbors requires more time to capture global information from the spectral perspective. Therefore, a dynamic-interactive subgraph Transformer (DISGT) is proposed for unsupervised hyperspectral BS. Specifically, a dynamic distance norm is designed to measure the correlation between bands and characterize several subspaces with better separability. Bands with uncertain subspace membership are shared with other subspaces through information interaction to enhance the diversity of band combinations, compensate for the effect of subspaces on global information processing, and construct subspaces into subgraphs. Furthermore, the subgraph Transformer (SGT) captures local and global spectral information using a sparse multihead attention (SparseMHA) mechanism to accurately evaluate band importance. Finally, the original HSI is reconstructed using the crucial bands to continuously adjust and obtain the optimal band subset. The superior performance of the DISGT is demonstrated through extensive comparisons with other state-of-the-art methods on three public datasets.**

***Index Terms*— Dynamic subspace partition (DSP), hyperspectral imagery (HSI), information interaction, subgraph Transformer (SGT), unsupervised band selection (BS).**

I. Introduction

HYPERSPECTRAL imagery (HSI) consists of dozens to hundreds of bands. The continuity of spectral attributes allows HSI to capture a wealth of information about the Earth's surface, providing sufficient features to accurately identify similar objects in classification tasks [1], [2], [3]. However, the multitude of bands also indicates an obvious increase in data dimensionality. There is a higher degree of similarity between adjacent bands, and the redundant features further increase the computational burden [4], [5]. Therefore, it is necessary to reduce the data dimensionality in HSI while minimizing the loss of classification accuracy as much as possible.

There are two key techniques for reducing data dimensionality: feature extraction and feature selection. Feature extraction involves mapping the original HSI to crucial features with informative and distinctive characteristics, aiming to reduce data dimensionality while providing more effective information for further analysis [6], [7], [8]. However, feature extraction transforms the data into a new feature space, resulting in a lack of physical meaning [9]. In contrast, feature selection involves directly choosing relevant features from the original HSI, preserving the intrinsic properties and improving the interpretability of models [10], [11], [12]. Feature selection is referred to as band selection (BS) in HSI processing and is broadly categorized into supervised and unsupervised methods. Since supervised methods heavily depend on annotated samples, which leads to expensive labeling efforts in practical applications [13], unsupervised methods concentrate on the data itself to determine band importance and decide whether to keep the current band, making them a hot research topic [14], [15], [16], [17].

In general, unsupervised BS methods are further categorized into three types: ranking [18], [19], searching [20], [21], and clustering [22], [23]. The ranking-based BS methods assess bands according to specific criteria and then select the crucial bands by top-rank [24], but it is difficult to determine the reasonable criteria for different datasets. The searching-based methods treat BS as a combinatorial optimization problem, obtaining suitable band subsets through heuristic information [25], but they are prone to falling into local optima. The clustering-based BS methods partition spectra into subspaces based on the regional difference and select the most representative bands from each subspace [26]. However, the correlation calculation in these methods depends on a static distance norm, which weakens their ability to recognize cluster structures and neglects the relationship between local information contained in subspaces.

The powerful end-to-end hierarchical representation capability and the advantages of nonlinear spectral analysis make the deep neural network (DNN) widely used in the field of

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HSI processing [27], [28]. Among them, the convolutional neural network (CNN) has become one of the most commonly used DNN models [29], [30]. For instance, the CNN with 1-D convolution was applied to BS based on distance–density (BSCNN+) to aggregate features and determine the band subset [31]. Furthermore, Li et al. [32] introduced a CNN-based BS method using an attention mechanism to capture nonlocal information and select important bands without deteriorating the classification. However, the coverage of the CNN is limited by the receptive field, which may oversimplify the similarity and combination between distant bands, ultimately reducing the classification accuracy of the final band subset.

To address the limitations of the CNN, spectra are constructed into a graph where bands are described as nodes and the correlations between bands are set as edges. Graph representation learning (GRL) is then put forward to fuse the node information, utilizing graphs to represent the flavors of different datasets, providing a suitable computational interface to deal with HSI [33], [34]. The graph neural network (GNN), as one of the most representative GRL models, typically combines with specific aggregation functions to analyze the graph structure information around the pivotal nodes [35]. The concept of convolution is transferred to non-Euclidean domains, yielding the graph convolutional network (GCN) [36], [37], [38], which propagates the connections between nodes and their neighborhoods layer by layer to represent the relationships of nodes in the graph [39]. However, these methods are difficult to quickly handle the dependencies of distant bands and capture global information due to the process of aggregating neighboring nodes multiple times.

In this article, a novel dynamic-interactive subgraph Transformer (DISGT) is proposed for unsupervised hyperspectral BS, which encompasses three main components: dynamic subspace partition (DSP), subspace information interaction (SII), and a subgraph Transformer (SGT). In particular, DSP with a dynamic distance norm is utilized to divide all bands into several subspaces. The membership degree is computed for bands to ensure their equality, and SII is performed to establish relationships between subspaces by sharing bands with uncertain subspace membership. After subspaces are constructed into subgraphs through SII, each subgraph corresponds to its own SGT in non-Euclidean domains, which is implemented to efficiently gain local information within subgraphs while capturing the global spectral information.

The main contributions of this article are listed as follows.

1) DISGT is proposed for unsupervised hyperspectral BS, which comprehensively takes both local and global spectral information into consideration from different HSI patches and retains crucial bands without affecting the classification accuracy.
2) DSP and SII are integrated to better distinguish the regional differences of spectra using a dynamic distance norm and to consider global information by establishing relationships between local information contained in subspaces.
3) SGT is designed to handle subgraphs with varying numbers of bands, enhancing the diversity of band combinations while reducing both time and space complexity in solving the BS problem.
4) The comprehensive experiments are performed on three public HSI datasets, and the experimental results demonstrate the superiority of the proposed DISGT over other state-of-the-art BS methods.

The subsequent organization of this article is as follows. Section II provides a brief overview of related work on BS. Section III presents the proposed unsupervised BS method, DISGT. Section IV reports the experimental results of the DISGT on three public datasets. Section V concludes this article, summarizing key findings and contributions of our proposed DISGT in the context of BS.

II. Related Work

In this section, various hyperspectral BS methods are reviewed, which can be categorized into three types: subspace-based, DNN-based, and GRL-based methods. We also discuss the motivation behind our proposed method.

A. Subspace-Based Band Selection

Subspace-based BS methods involve grouping bands with high similarity into the same subspace and then selecting the corresponding bands from each subspace separately. Yao et al. [40] proposed an end-to-end HSI change detection network based on BS (ECDBS), which employed diffusion operations to reduce differences in relationships between subspaces and facilitated learning of band weights for BS. Wang et al. [41] presented a BS method with an adaptive subspace partition strategy (ASPS), which partitioned spectra by maximizing the ratio of interclass to intraclass distances, and selected the least noisy bands as the representatives, considering the order and intergroup analyses. In addition, Wang et al. [42] proposed a BS method with a fast neighborhood grouping strategy (FNGBS), which selected bands based on maximum local density and information entropy in subspaces, and recorded distances between bands and subspace center to create a similarity matrix for final partitioning. Wang et al. [43] introduced a spatial–spectral graph regularization (GRSC), which incorporated spatial information and regional differences, generating discriminative latent features through superpixel segmentation to explore spectral characteristics via self-representation subspace clustering. However, these methods are mainly based on static distance norms to measure the correlations between bands, which limits their ability to accurately identify cluster structures. Furthermore, as the number of bands increases, the size of subspaces decreases, which is not conducive to contacting distant bands.

B. DNN-Based Band Selection

DNNs have received significant attention in HSI processing due to their powerful nonlinear learning and abundant data representation capabilities. Among them, the CNN with local receptive fields and shared weights is prevalent. Esmaeili et al. [44] embedded 3-D convolutional layers into a genetic algorithm to identify suitable or optimal band subsets

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and the 3-D CNN was implemented as the objective function and offered new insights for BS. Cai et al. [45] introduced an end-to-end BS method (BS-Net), which consisted of a band attention mechanism and a reconstruction network, the former used the CNN to measure the correlation between bands, and the latter reconstructed the original HSI using the selected bands. Roy et al. [46] proposed an end-to-end unsupervised dual-attention reconstruction network for BS (DARecNet), which employed two attention modules with position and channel and multidimensional CNNs to recalibrate feature maps, extracting nonlinear contextual information both spectrally and spatially. Feng et al. [47] introduced a BS method based on bandwise-independent convolution and hard thresholding (BHCNN), which applied 1×1 convolutions for each band and constructed a multiscale 3-D dilated convolution for joint spatial–spectral feature extraction from the selected bands. However, the above methods primarily rely on local receptive fields, which limits their ability to capture global spectral information. This limitation adversely affects the measurement of correlations between bands, especially for distant bands, which is crucial for enhancing the diversity of band combinations.

C. GRL-Based Band Selection

GCN is a commonly used model in GRL, where correlations between bands are set as edges, and the constructed graphs are processed through the GCN to overcome the limitations of the CNN with fixed receptive fields. Tang et al. [48] presented a self-representation model with spatial and spectral structure for unsupervised hyperspectral BS (S^4P), which incorporated an adaptive and weighted graph fusion term to generate a unified similarity graph across different regions. Feng et al. [49] introduced a dual GCN (BSD-GCN), which constructed both band-based and sample-based graphs, using band attention mechanisms and sparse constraints to filter and assess band weights. Yu et al. [50] proposed a marginalized graph self-representation approach for BS (MGSR) that formed different homogeneous regions in HSI through superpixel segmentation and utilized the GCN to extract relationships between adjacent pixels, thereby improving the robustness of BS. Feng et al. [53] introduced a novel zero-shot method (MR-Selection), utilizing meta-reinforcement learning and a dynamic structure-aware GCN to effectively capture complex correlations between bands, along with a two-stage optimization strategy to enhance selection efficiency and model generalization. However, the differences in heterogeneous bands are usually obscure from the spectral perspective, making it difficult to exploit new band combinations and obtain more suitable band subsets. Moreover, the multilayer aggregation of the GCN, which extracts global information from the spectral perspective, increases both the time and space complexity of BS.

D. Motivation

The above BS methods based on subspace and the DNN are primarily adapted to Euclidean metrics. Among them, subspace-based BS methods utilize a static distance norm to measure the correlations between bands, resulting in weak generalization ability across different HSI datasets [51]. In addition, DNN-based BS methods implement convolutions to focus on the entire spectral dimension, making it challenging to be concerned with the similarity of distant bands [52]. Although bands are treated as nodes to help GRL-based BS methods overcome the limitations imposed by fixed receptive fields, these methods are still accompanied by a higher computational burden and neglect the regional differences of spectral structure.

To leverage the advantages of mutual compensation between GRL and subspace, the DISGT is proposed to update the band combinations by partitioning spectra into several subspaces, establishing relationships between these subspaces, constructing subspaces into subgraphs, and designing SGT to effectively capture both local and global spectral information. This ensures that crucial bands are preserved, maintaining classification accuracy without the need for any label information. Moreover, the band combinations are reconstructed to the original HSI, and the optimal band subset is obtained through the convergence of the DISGT.

III. Proposed Method

The original 3-D HSI tensor is given as $X \in \mathbb{R}^{W\times H\times B}$, where W, H, and B represent the length and width of the image as well as the number of bands, respectively. X is cut into N uniform HSI patches. After flattening, each patch is transformed from a 3-D tensor into a 2-D spectral matrix, which is defined as $x_p \in \mathbb{R}^{B\times M}$ $(M = m \times m)$, where $1 \leq p \leq N$, and m is the length and width of the pth patch.

As shown in Fig. 1, the proposed DISGT is encompassed into four components: the DSP module, the SII module, the SGT module, and a reconstruction block. The DSP module dynamically allocates heterogeneous bands from each patch to excavate regional differences in spectral structure and compute the membership matrix. The SII module shares the band with fuzzy information to realize the establishment of relationships between subspaces and constructs subspaces into subgraphs. The SGT module filters local and global information from the interacted subgraphs and assigns weights to the crucial bands. Finally, the reconstruction block reconstructs the crucial bands and compares them with the original HSI patches to train the DISGT. The crucial bands in each patch are accumulated to obtain the band subset of the entire HSI, achieving an end-to-end unsupervised BS method. In this section, the four main parts of the proposed DISGT are introduced in detail.

A. Dynamic Subspace Partition

In subspace-based BS methods, the static distance norm is applied to measure the correlation between bands, which decreases their ability to identify cluster structures and makes it difficult to better distinguish the similarities and differences between bands. The basic idea of DSP is to analyze the properties of each band through dynamic distance norm and allocating bands with similar attributes to the same subspace.

Bands are regarded as training samples, and the sample matrix is defined as $T = \{t_1, t_2, \ldots, t_i, \ldots, t_B\} \in \mathbb{R}^{B\times M}$,

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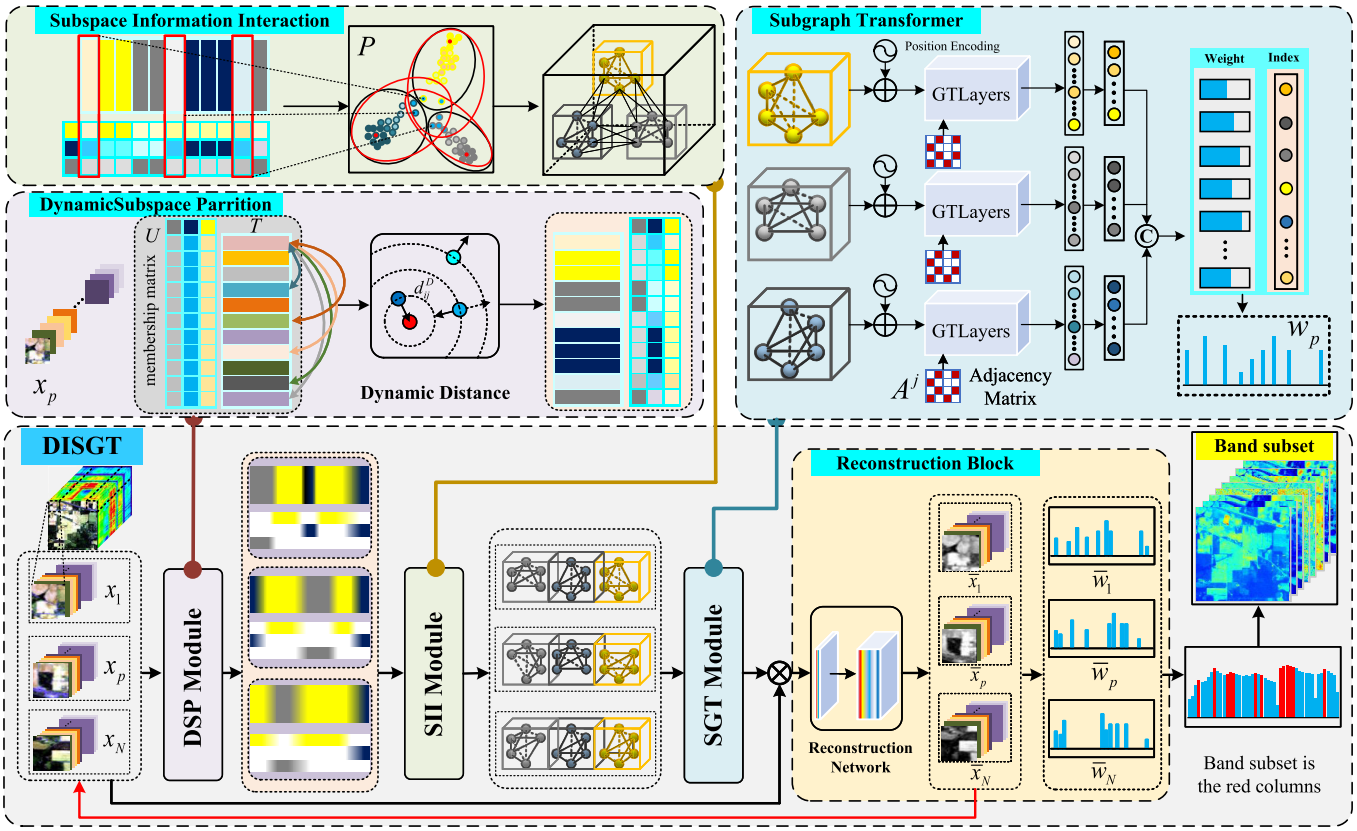


Fig. 1. Architecture of the DISGT. The symbol ⊗ represents element-wise multiplication between matrices, ⊕ represents element-wise addition between matrices, and © represents concatenation between vectors.

and $t_i \in \mathbb{R}^{1\times M}$ is the ith band vector. The number of subspaces is set to S, and the membership matrix is defined as $U \in \mathbb{R}^{S\times B}$. The subspace center matrix is defined as $C = \{c_1, c_2, \ldots, c_j, \ldots, c_S\} \in \mathbb{R}^{S\times M}$, and $c_j \in \mathbb{R}^{1\times M}$ is the jth subspace center vector. The dynamic distance norm between bands and subspace centers is defined by the following equations:

$$d_{ij}^D = \max\left\{||t_i - c_j||_2^2 + \alpha_{ij}||t_i - c_j||_2, 0\right\} \tag{1}$$

$$\alpha_{ij} = \begin{cases} \alpha, u_{ij} > \partial \\ -\alpha, u_{ij} \le \partial \end{cases} \tag{2}$$

where d_{ij}^D represents the similarity of the ith band and the jth center, respectively, with $1 \le i \le B$ and $1 \le j \le S$. u_{ij} denotes the membership degree between the ith band and the jth center, ∂ is the membership threshold, $0 \le \partial \le 1$, which means that the distribution of bands with strong correlation changes dynamically with the dataset. The parameter α is regarded as the amplitude coefficient for dynamic calibration, $0 \le \alpha \le 1$. The difference between using dynamic and static distances to calculate the correlation between bands and subspace centers is shown in Fig. 2.

As shown in Fig. 2, the distributions of membership in dynamic and static distance norms exhibit obvious differences. The membership of the dynamic distance norm is larger and more uniform than that of the static distance norm in the central area of the subspace. When the membership between the band and subspace center is greater than ∂, d_{ij}^D will make them closer to each other, making such bands more reliable. A larger ∂ will cause fewer reliable bands, which generates more redundant information for this subspace after information interaction. Conversely, a smaller ∂ will share fewer bands with a fuzzy membership degree, decreasing the diversity of band combinations.

With the addition of dynamic distance norms, the subspace centers and membership of each band need to be updated. The alternating minimization criterion is used to provide suitable conditions for stopping the update as follows:

$$\begin{cases} \min\limits_{U,C} J_{\mathrm{DSP}}(U,C) = \sum\limits_{i=1}^{B}\sum\limits_{j=1}^{S} u_{ij}^2 d_{ij}^D \\ \quad + \sum\limits_{i=1}^{B}\sum\limits_{j=1}^{S} u_{ij}^2 \sum\limits_{r\in N_i, r\ne i} \dfrac{(1-u_{rj})^2 d_{rj}^D}{d_{ir}+1} \\ \text{s.t.} 0 \le u_{ij} \le 1, \sum\limits_{j=1}^{S} u_{ij} = 1,\ 0 < \sum\limits_{i=1}^{B} u_{ij} < n \end{cases} \tag{3}$$

where $U \in \mathbb{R}^{S\times B}$ is the membership matrix and N_i is the neighborhood of the current band. The position constraint of bands is improved by joining its neighbors, which boosts the ability of subspace partition. By adding (1)–(3) and taking the derivative of u_{ij} and c_j, it is revealed that the membership of bands and subspaces constrain each other to reach the stable state. The dynamic distance norm can acquire more uniform areas with better separability than other subspace partition strategies, facilitating the sharing of more bands with fuzzy information to enhance the diversity of band combinations.

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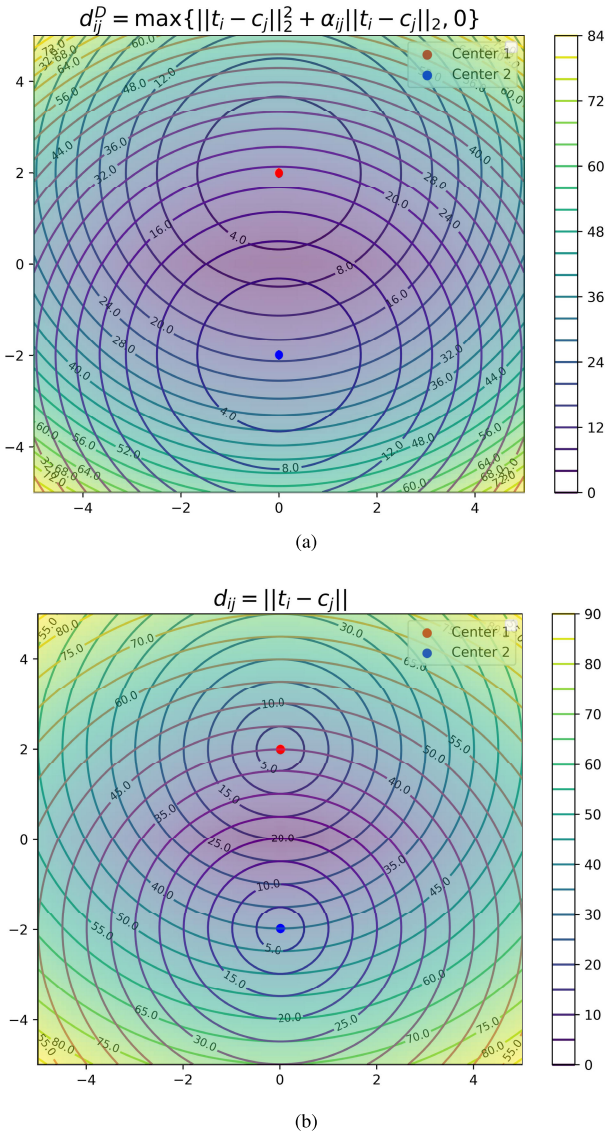


Fig. 2. Contour maps of membership of the (a) dynamic distance norm and (b) static distance norm for two subspaces.

B. Subspace Information Interaction

Partitioning the subspace without considering the relationships between them makes it difficult to capture the global spectral information. The basic idea of SII is to establish the relationship between different subspaces by sharing the bands with fuzzy membership degrees and reflect the correlation of distant and neighboring homogeneous bands by constructing interacted subspaces into subgraphs.

The membership matrix in DSP is used as the input for SII. The interacted subspace set is defined as $P = \{p_1, p_2, \ldots, p_j, \ldots, p_S\}$, where p_j represents the band index subset for the jth subspace and $||p_j||$ is the number of bands in the jth subspace. It should be noted that $||p_1||$ may not be equal to $||p_2||$ or $||p_j||$, therefore, P is not a matrix. The interaction process is defined by the following equation:

$$\begin{cases} p_j = \{i | u_{ij} \geq \varphi,\ 0 \leq i < B\} \\ \text{s.t.} \varphi = \dfrac{1}{S},\ \displaystyle\sum_{j=1}^{S} ||p_j|| > B \end{cases} \tag{4}$$

where φ is the interaction threshold. Bands whose membership degree is greater than φ are selected from U and added into the corresponding subspace. Since the sum of membership degrees of each band in different subspaces is 1, if φ is set to greater than $(1/S)$, each band is assigned to a unique subspace or not assigned to any subspaces. The size of φ is inversely proportional to the amount of information interaction.

Taking three subspaces as an example in Fig. 1, there is a noticeable intersection between the new subspaces compared to DSP, which consists of the bands with fuzzy membership degrees. The diversity of band combinations is enhanced by sharing bands with uncertain membership into other subspaces from both subspace and spectral perspectives.

To further improve the correlation between distant and neighboring bands and realize flexible capturing of global spectra information and dynamic weight adjustment, the interacted subspaces are constructed into subgraphs. The subgraph matrix $g_j \in \mathbb{R}^{||p_j|| \times M}$ is formed by selecting the bands from p_j out of T, the subgraph set is defined as $G = \{g_1, g_2, \ldots, g_j, \ldots, g_S\}$, and the adjacency matrix of the jth subgraph is defined as $A^j \in \mathbb{R}^{||p_j|| \times ||p_j||}$. Consequently, the jth subspace corresponds to a band index subset p_j, a subgraph matrix g_j, and an adjacency matrix A^j. Since it is difficult to distinguish the position between bands from the spectral perspective in the non-Euclidean domain, the relative position of bands is incorporated to calculate the weight of the edge between two nodes, which is defined by the following equation:

$$A_{ik}^{j} = \begin{cases} ||t_i - t_k||^2 \cdot \exp(-|i-k|), & \text{if } i \neq k \\ 0, & \text{otherwise} \end{cases} \tag{5}$$

where A_{ik}^{j} represents the weight between the ith node (band) and the kth node (band) in the jth subgraph. When these nodes are closer together from the spectral perspective, A_{ik}^{j} is closer to the square of their Euclidean distance.

C. Subgraph Transformer

After constructing each subspace into a subgraph, the fundamental idea of the SGT is to quickly search for crucial bands by aggregating the local information through the GCN and implementing a sparse multihead attention (SparseMHA) mechanism to effectively extract global spectral information. The structure of the SGT is shown in Fig. 3.

The SGT assigns a corresponding Graph Transformer to each subgraph, and each Graph Transformer is composed of multiple graph transformer layers (GTLayers). First, the subgraph matrix is disposed of Laplacian position encoding (PE), GCN is used to aggregate local information for each node, and this process is called feature extracting (FE). The output of PE is transformed to the same dimension as the output of FE, and then they are fused to generate a subgraph attention matrix $h_j \in \mathbb{R}^{||p_j|| \times d}$, which is defined by the following equation:

$$h_j = \text{PE}(g_j) \oplus \text{FE}(g_j) \tag{6}$$

where d is the dimension of aggregated features and $\oplus$ represents element-wise addition between matrices. Furthermore,

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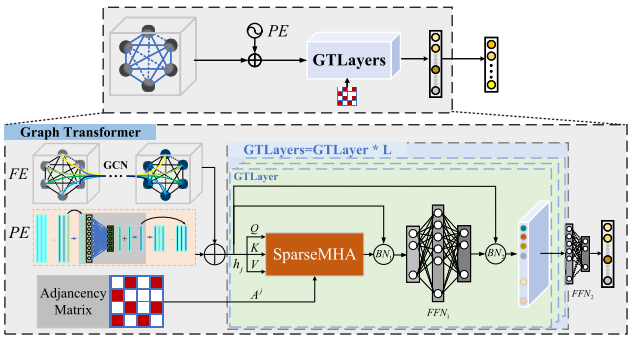


Fig. 3. Structure of the SGT. PE is the Laplacian position encoding operation and FE is the feature extracting operation using the GCN.

input subgraph attention matrix h_j and adjacency matrix A^j to the SparseMHA. The query, index, and value matrices are defined as $Q, K, V \in \mathbb{R}^{||p_j|| \times d}$, respectively. The SparseMHA is defined by the following equation:

$$\text{SparseMHA}(Q, K, V, A) = \text{softmax}\left(\frac{(QK^T) \circ A^j}{\sqrt{d}}\right)V. \tag{7}$$

After the query and index matrices are blended, they are element-wise multiplied with the adjacency matrix. This result is then multiplied with the value matrix to produce the updated subgraph attention matrix, which is subsequently input to the first feed-forward network (FFN_1) and two batch normalization (BN) layers. This process constitutes a GTLayer.

The detailed update process of the GTLayer is defined by (8)–(12)

$$\dot{h}_j^{\ell+1} = \left(\mathop{||}_{k=1}^{H}\left(W^{k,\ell} h_j^{\ell} V^{k,\ell}\right)\right) O^{\ell} \tag{8}$$

$$W^{k,\ell} = \text{softmax}\left(\frac{h_j^{\ell} Q^{k,\ell} \cdot \left(h_j^{\ell} K^{k,\ell}\right)^T}{\sqrt{d_k}} \cdot A^j\right) \tag{9}$$

where H is the number of attention heads, A^j is the adjacency matrix of the jth subgraph, d_k is the dimension of each attention head, and $d = d_k \times H$. The subgraph attention matrix of the ℓth layer in the jth subgraph is ${h_j}^{\ell} \in \mathbb{R}^{||p_j|| \times d}$. $Q^{k,\ell}, K^{k,\ell}, V^{k,\ell} \in \mathbb{R}^{d \times d_k}$ are the query, index, and value matrices in the kth attention head of the ℓth layer, respectively. After the softmax operation, the attention weight matrix $W^{k,\ell} \in \mathbb{R}^{||p_j|| \times ||p_j||}$ in the kth attention head is derived. The symbol $||$ denotes the concatenation of outputs from all attention heads, and $O^{\ell} \in \mathbb{R}^{d \times d}$ is the output matrix of SparseMHA. After SparseMHA, the updated subgraph attention matrix ${\dot{h}^{\ell+1}}_s$ is input to the FNN_1 and processed through a residual connection and BN. The specific process of residual connection is defined by the following equations:

$${\ddot{h}_j}^{\ell+1} = \text{BN}_1\left({h_j}^{\ell} + {\dot{h}_j}^{\ell+1}\right) \tag{10}$$

$${\dddot{h}_j}^{\ell+1} = \text{FFN}_1\left({\ddot{h}_j}^{\ell+1}\right) \tag{11}$$

$${h_j}^{\ell+1} = \text{BN}_2\left({\ddot{h}_j}^{\ell+1} + {\dddot{h}_j}^{\ell+1}\right) \tag{12}$$

where ${h_j}^{\ell+1}$ is the subgraph attention matrix of the $\ell+1$th layer in the jth subgraph. After training of all GTLayers, the final subgraph attention matrix h_j^L is mapped to a score vector $\vec{s_j} \in \mathbb{R}^{||p_j|| \times 1}$ through the second feed-forward network (FFN_2) with the following equation:

$$\vec{s_j} = \text{FFN}_2\left(h_j^L\right). \tag{13}$$

The top-ranked nodes are selected within each subgraph, and all subgraphs are connected via the SGT to derive the sparse mask vector $w_p \in \mathbb{R}^{B \times 1}$ for the pth HSI patch, as defined in the following equation:

$$\begin{cases} w_p = \mathop{||}_{j}^{S} f_{\text{top}}\left(\vec{s_j}, b_j\right) \\ \text{s.t.} \sum_{j=1}^{S} b_j = b \end{cases} \tag{14}$$

where b_j denotes the number of selected nodes in the jth subgraph and b represents the number of selected bands. Specifically, b_j is determined based on the proportion of bands in each subgraph relative to the total number of bands. The function $f_{\text{top}}(\cdot)$ represents the operation of sorting $\vec{s_j}$ in descending order and selecting the top b_j values. The selected elements are retained by their original value, while other elements are set to 0. All selected bands from the subgraphs are concatenated to form the sparse mask vector w_p.

D. Reconstruction Blocks

To evaluate the quality of the selected bands in the SGT, a reconstruction block is designed to update the SGT module and achieve its optimal state.

A set is formed by the SGT using the top-ranked bands in x_p through w_p. The core of this block is to reconstruct the original HSI patch using a minimal number of bands, aiming to minimize the loss of classification accuracy. w_p serves as a mask to reconstruct the HSI patch $\bar{x}_p$, as shown in the following equation:

$$\bar{x}_p = \text{Rec}\left(x_p \otimes \left(w_p \cdot r\right)\right) \tag{15}$$

where $r \in \mathbb{R}^{1 \times M}$ is a row vector with all elements equal to 1, $\otimes$ represents the element-wise multiplication, and $\text{Rec}(\cdot)$ is the reconstruction network implemented through a feed-forward network. Finally, the loss between $\bar{x}_p$ and x_p is calculated to adjust the SGT module and select better bands in the next patch.

The final sparse mask vector $\bar{w}_p \in \mathbb{R}^{B \times 1}$ is derived by completing the training of the SGT for each patch. The weights of corresponding bands in each patch are accumulated to evaluate the band importance across the entire HSI. The top-ranked bands are then selected to form the band subset, as described in the following equation:

$$\text{Bs} = \arg\max\left(\sum_{p=1}^{N} \bar{w}_p, b\right) \tag{16}$$

where $\text{Bs} \in \mathbb{R}^{b \times 1}$ is the final band subset output by the DISGT.

The implementation details of the DISGT are shown in Algorithm 1.

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Algorithm 1 DISGT

Input: HSI cube: X; the band subset size: b; the number of subgraphs: S, and DISGT hyper-parameters.
Output: Band subset: Bs.

Process X to N patches.
for $p = 1, 2, \ldots, N$ **do**
  Calculate the dynamic distance d^D between bands and subspace centers by Eqs. (1) and (2).
  Update the membership matrix U and subspace center matrix C by Eq. (3).
  Share the bands of fuzzy membership degree to other subspaces by Eq. (4).
  for $j = 1, 2, \ldots, S$ **do**
    Calculate adjacency matrices A^j of subgraphs by Eq. (5).
    Update subgraph attention matrix h_j^ℓ by Eqs. (8)-(12).
  end for
  Calculate the sparse mask vector w_p by Eq. (14).
  Reconstruct HSI patch by Eq. (15).
  Calculate the loss between the reconstructed HSI patch and the original HSI patch.
end for
Accumulate sparse attention vector of each patch and select the top-ranked bands to form the final band subset Bs by Eq. (16).

IV. Experiments

In this section, the experiments are conducted to assess the effectiveness of the proposed DISGT. First, three commonly used public HSI datasets are introduced in Section IV-A. Furthermore, the experimental setups involving the comparative methods, evaluation metrics, and parameter settings are introduced in Section IV-B. Sections IV-C–IV-E provide the comparisons of the proposed unsupervised BS with ten representative methods. To further investigate how the performance is improved by using our method, an ablation study is presented in Section IV-F. Finally, the running time of different methods for selecting a certain number of bands is compared in Section IV-G.

A. Datasets

The performance of the proposed method is evaluated on three public datasets: the Indian Pines, Pavia University, and Hanchuan datasets. Table I lists their details.

1) Indian Pines: The scene was collected by the AVIRIS imaging spectrometer in a piece of Indian Pines in Indiana, USA, with a spatial resolution of 20 m, consists of 145 × 145 pixels, and 224 spectral bands from 0.4 to 2.5 μm. There are 16 classes of land-cover objects in the scene. Since bands [104–108], [150–163], and 220 correspond to water absorption, a total of 20 bands were removed, retaining 200 spectral bands.

2) Pavia University: The scene was collected by the reflective optics system imaging spectrometer (ROSIS) sensor and recorded a scene over Pavia. It consists of 610 × 340 pixels and 115 bands in the original dataset and the spatial resolution is 1.3 m. By removing 12 noisy bands, 103 spectral bands are retained to describe nine classes in this extracted dataset.

3) Hanchuan: The scene was collected by the Headwall Nano Hyperspectral imaging spectrometer aboard the drone platform in Hanchuan, Hubei Province, China, with a spatial resolution of about 0.0109 m. The image has 274 bands, and 1217 × 303 pixels are retained to describe 16 different classes of land cover.

TABLE I
Summary of Indian Pines, Pavia University, and Hanchuan Datasets

Data sets	Indian Pines	Pavia University	Hanchuan
Pixels	145 × 145	610 × 340	1217 × 303
Channels	200	103	274
Classes	16	9	16
Sensors	AVIRIS	ROSIS	Headwall Nano

TABLE II
Hyperparameter Settings for Different BS Methods

Comparative methods	Indian Pines	Pavia University	Hanchuan
ASPS	B=3, M=10	B=10, M=10	B=10, M=10
FNGBS	k=2, Z=10	k=3, Z=1	k=2, Z=10
GRSC	α=1, β=1, d=5, k=15		
BS-Net	λ=1e-2, η=2e-3, maxiter=100		
DARecNet	η=1e-2, maxiter=100		
BHCNN	N=15, λ=0.3, T=1200		
BSD-GCN	η=1e-3, T=1000		
MGSR	λ=100, p=0.9, S=25		λ=100, p=0.9, S=200
MR-Selection	α=-0.0001, β=0.01		
S^{4}P	α=100, γ=0.01, T=2000		

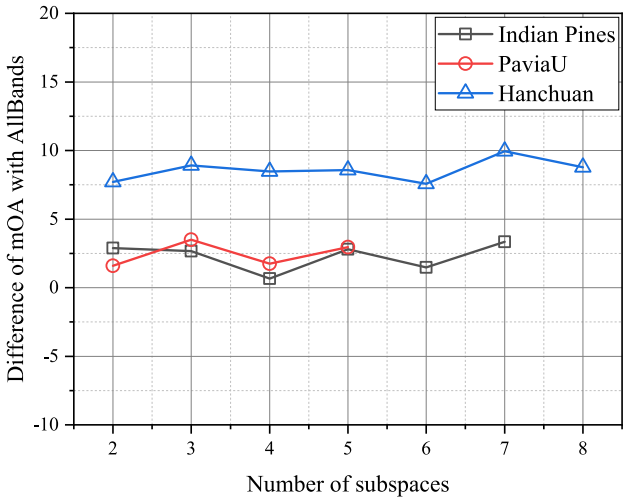


Fig. 4. mOA differences between AllBands and DISGT in 5–30 bands with different numbers of subgraphs on three datasets.

B. Experimental Settings

1) Comparative Methods: To verify the performance of the DISGT, ten representative BS methods are utilized for comparison on the same dataset, including subspace-based methods ASPS [41], FNGBS [42], and GRSC [43], DNN-based methods BS-Nets [45], DARecNet [46], and BHCNN [47], and GRL-based methods BSD-GCN [49], MGSR [50], MR-Selection [53], and S^{4}P [48]. In addition, AllBands is taken as an important reference by using all bands of the HSI dataset. To ensure the reliability of the experiment, each method is performed 20 times. Table II lists the parameter settings of all comparison tests.

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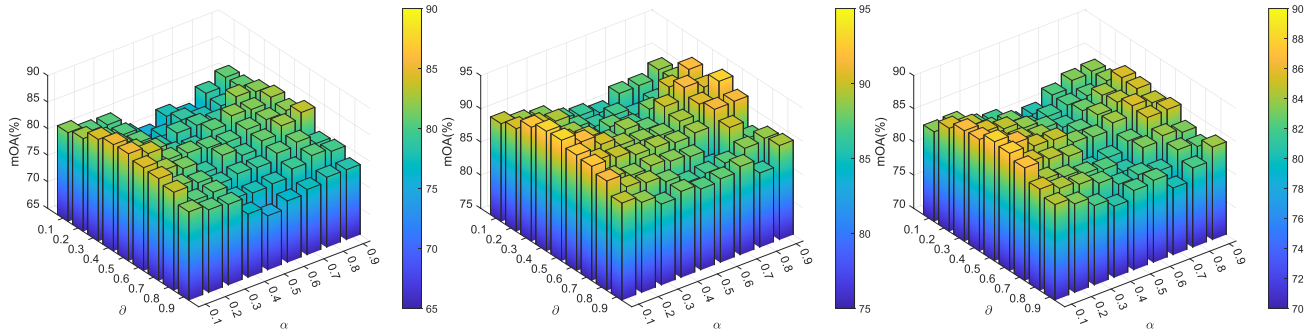


Fig. 5. Sensitivity analysis of parameters on the three datasets. (a) Indian Pines. (b) Pavia University. (c) Hanchuan.

2) Evaluation Metrics: The classification accuracy of the final band subset is taken as its quality judgment. Three typical classifiers—support vector machines (SVMs) [54] with a radial kernel function, k-nearest neighbor (kNN) [55] with $k = 5$, and random forest (RF) [56] with 500 trees—are used to compare the classification accuracy with the optimal band subsets. For the fairness of the experiment, 10% of samples from each class are randomly selected as the training set and the remaining samples are acted as the testing set. Three commonly used indices—overall accuracy (OA), average accuracy (AA), and Kappa coefficient (Kappa)—are employed to quantify the effectiveness of different methods. The proposed DISGT is implemented in the PyTorch framework using the Python language, and all the experiments are performed on a computer equipped with an Intel Xeon E-2286M CPU and an Nvidia RTX 5000 GPU.

3) Parameter Configurations: The hyperparameters are tuned using various approaches. Since too many subgraphs may result in some of them without any bands, it is necessary to decide the number of subgraphs S. According to the total number of bands, S is set to 2–7, 2–5, and 2–8 for three datasets, respectively. Fig. 4 demonstrates the differences in mean OA (mOA) obtained by the DISGT and AllBands for ten independent operations with 5–30 bands and different numbers of subgraphs.

It is observed that the mOA with four subgraphs is optimal for the Indian Pines dataset. For the other two datasets, the number of subgraphs is set to 2 and 6, respectively, to achieve the highest mOA. To improve the performance of the DSP module, Fig. 5 shows the variation in mOA over 5–30 bands for different datasets as the values of affiliation threshold ∂ and dynamic calibration coefficient α are uniformly varied from 0.1 to 0.9.

As shown in Fig. 5, the mOA over 5–30 bands at different ∂ and α values shows a regular change. When α is fixed, both too-small and too-large values of ∂ will reduce mOA. When ∂ is fixed, both oversized and undersized values of α will improve mOA. This is because a larger ∂ makes it more difficult to explore bands with strong correlations. In contrast, a smaller ∂ makes it difficult to distinguish the correlations between bands. The dynamic calibration coefficient α can remedy the shortcomings of an unsuitable ∂. A larger α helps ∂ improve the correlation of distant bands, while a smaller α slows down the formation of weak correlations between bands. Through comparative trials, ∂ and α are set to 0.5 and 0.1, respectively.

C. Results on the Indian Pines Dataset

In this experiment, the performance of the DISGT is evaluated on the Indian Pines dataset. Fig. 6(a)–(c) shows the OA obtained by different BS methods and AllBands under three classifiers. The size of the band subset b is set from 5 to 30, with an interval of 2. To ensure the reliability of the classification accuracy, each method is performed 20 times.

From Fig. 6(a)–(c), the proposed DISGT consistently delivers superior OA across the SVM, kNN, and RF in most cases. In particular, the DISGT outperforms other methods in classification accuracy for all classifiers when the band subset size exceeds 13. In addition, the DISGT achieves a higher OA than AllBands for SVMs when b is more than 21, whereas other methods require more bands to approximate the grade of AllBands. Notably, more bands do not always improve the classification accuracy in Fig. 5(a). For example, GRSC and BHCNN show a decreasing trend in OA when b is greater than 21. However, the DISGT can better distinguish crucial bands and avoid the Hughes phenomenon.

To quantitatively analyze different methods, Table III lists a detailed comparison of classification accuracy for 25 bands by the SVM. As can be seen in Table III, the proposed DISGT outperforms other comparative methods and achieves higher classification accuracy across ten classes. Fig. 7 visualizes the distribution of band subsets from different methods and the entropy value of each band. The band subsets of the DISGT, BS-Net, FNGBS, MR-Selection, S^4P, and ASPS possess a more uniform distribution compared to those from MGSR, BSD-GCN, BHCNN, DARecNet, and GRSC. This uniformity indicates that the DISGT is more effective in capturing local spectral information and integrating neighboring bands. Compared to the DISGT, the entropy values of the bands selected by BS-Net, FNGBS, MR-Selection, S^4P, and ASPS are insufficient. Fig. 8 is the classification map for different BS methods using 25 bands with the SVM, and the proposed DISGT provided better results, especially in yellow, blue, and purple regions.

D. Results on the Pavia University Dataset

In this section, the classification accuracy of DISGT is compared with ten representative methods on the Pavia University

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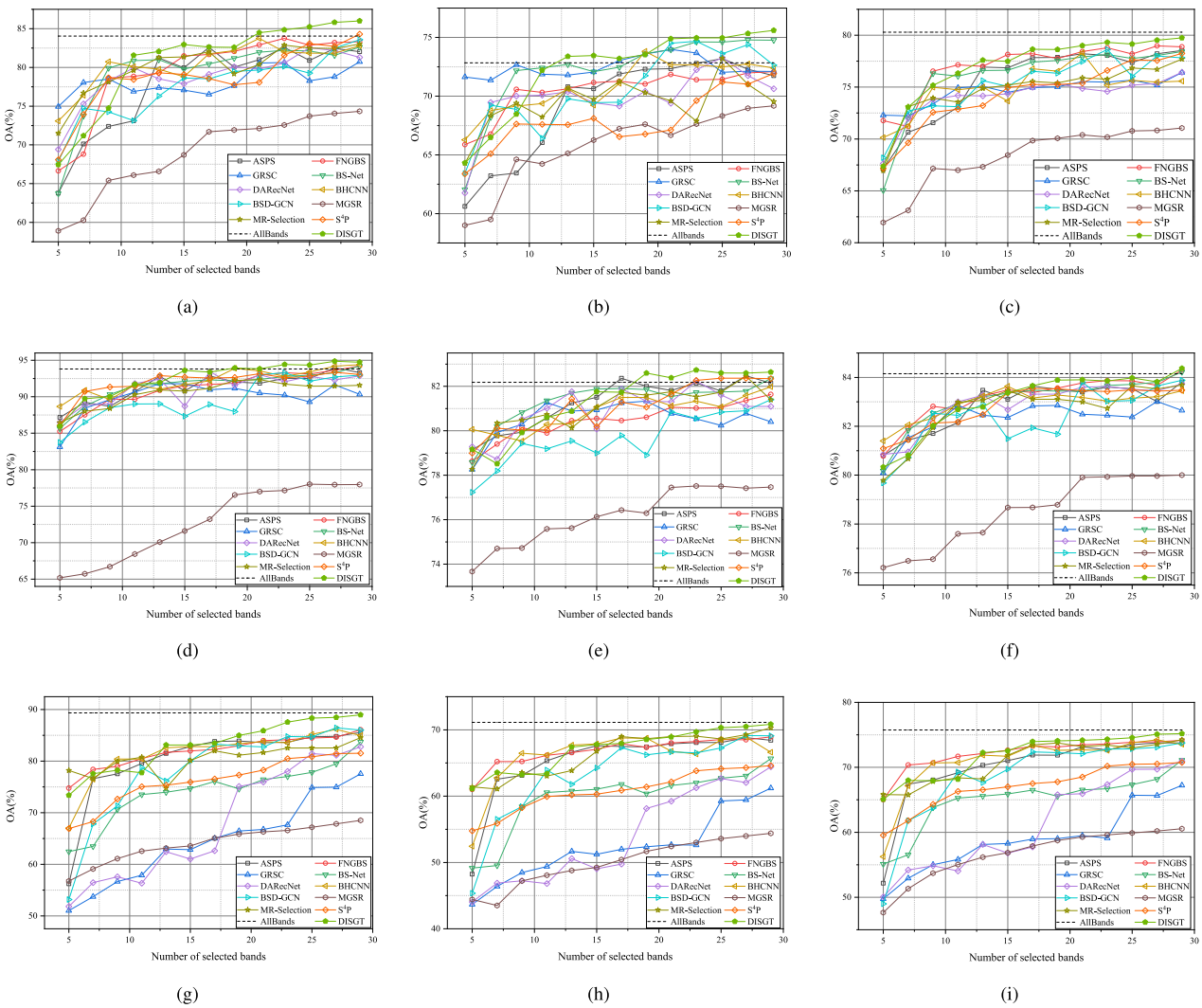


Fig. 6. Classification performance of 11 BS methods under different band subset sizes. (a)–(c) Classification accuracy on the Indian Pines dataset using the SVM, KNN, and RF as classifiers. (d)–(f) Classification accuracy on the Pavia University dataset using the SVM, KNN, and RF as classifiers. (g)–(i) Classification accuracy on the Hanchuan dataset using the SVM, KNN, and RF as classifiers.

TABLE III

CLASSIFICATION ACCURACY OF DIFFERENT BS METHODS USING 25 BANDS ON THE INDIAN PINES DATASET. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

No.	AllBands	ASPS	FNGBS	GRSC	BS-Net	DARecNet	BHCNN	BSD-GCN	MGSR	MR-Selection	S^{4}P	DISGT
1	64.61±23.43	68.23±20.86	70.89±12.81	57.42±27.45	72.23±7.81	77.01±5.12	78.61±8.47	76.41±20.65	55.83±21.81	61.16±4.91	69.91±7.64	**79.21±14.08**
2	78.63±3.42	70.61±3.68	73.23±0.87	64.21±0.88	74.25±2.81	77.84±1.86	75.21±5.28	70.47±2.45	65.07±2.12	73.59±0.82	74.80±3.17	**79.68±3.41**
3	78.45±2.63	75.04±3.11	78.67±1.64	70.13±1.07	75.84±5.22	**82.51±3.12**	76.87±3.24	74.04±4.93	67.41±4.68	76.68±0.74	74.45±1.51	78.71±5.03
4	66.04±6.12	65.61±6.42	70.64±6.46	56.27±3.24	66.63±3.62	68.02±9.18	61.82±4.84	65.01±7.06	50.46±5.45	71.66±3.08	68.03±1.63	**72.84±6.27**
5	91.21±2.84	84.23±2.21	86.87±5.81	89.42±1.72	89.41±3.62	87.27±3.23	91.64±2.01	84.22±2.81	86.47±3.64	89.03±1.46	87.76±2.49	**91.67±2.61**
6	91.12±4.96	91.21±1.26	90.56±1.23	88.64±2.76	89.64±2.61	89.45±1.66	90.81±3.87	89.24±2.23	86.83±2.21	88.94±1.77	91.72±0.91	**93.27±1.82**
7	90.24±9.87	86.21±24.23	**97.24±4.26**	76.07±14.23	75.47±16.61	92.07±6.12	81.87±18.22	73.41±26.67	69.83±16.07	83.75±16.25	94.58±7.72	86.42±13.47
8	92.81±2.21	93.87±1.84	96.07±3.37	92.81±3.28	95.81±2.83	95.87±1.21	**97.65±2.63**	93.67±2.46	92.42±1.67	94.41±1.67	97.40±0.11	96.87±1.81
9	71.17±18.01	63.67±5.61	**88.67±15.61**	69.48±20.61	67.48±25.42	66.37±14.01	69.04±14.13	72.24±14.11	56.67±15.72	74.58±11.99	73.33±9.67	57.41±20.67
10	79.01±2.89	75.25±2.24	78.47±3.61	76.87±3.23	**82.67±3.64**	78.81±3.23	76.07±1.21	72.67±2.61	64.45±1.64	80.93±1.98	81.03±3.36	81.65±2.67
11	83.04±2.78	79.21±1.23	81.81±1.82	75.47±1.42	81.07±1.45	79.47±1.10	79.04±1.29	79.09±2.11	66.47±2.41	82.51±1.61	82.28±1.34	**84.07±1.05**
12	82.22±3.81	82.21±3.04	84.01±2.91	75.84±3.27	81.41±2.45	84.23±2.81	85.44±4.61	81.63±2.68	74.41±3.68	86.55±1.33	87.55±2.19	**87.84±3.25**
13	93.07±5.77	93.41±4.41	92.87±3.81	93.67±4.45	90.61±3.45	93.82±3.24	95.04±4.08	90.61±4.44	88.42±7.64	90.36±5.19	88.05±0.51	**95.67±3.42**
14	93.62±3.64	92.25±0.87	93.25±0.84	92.81±2.25	94.01±1.37	91.65±2.63	93.24±1.28	92.45±2.44	92.42±2.44	93.45±1.33	**94.57±0.84**	94.48±1.47
15	75.07±4.36	72.24±4.24	70.07±2.87	70.24±3.25	69.85±5.87	70.84±1.23	74.22±4.87	65.04±5.21	61.87±6.24	69.56±3.11	69.62±2.95	**78.01±7.56**
16	99.41±0.68	98.01±2.97	98.06±3.11	98.25±2.27	97.64±1.61	98.45±3.41	96.47±2.64	96.25±4.26	93.64±4.67	95.07±2.68	96.95±1.77	**98.67±1.63**
OA(%)	83.91±1.24	80.41±1.45	82.62±0.45	78.14±0.87	82.32±0.33	82.81±0.53	82.21±0.66	79.67±0.81	73.57±0.81	83.02±0.17	83.19±0.77	**85.61±0.65**
AA(%)	83.16±0.84	80.67±2.61	84.35±2.82	77.49±1.24	81.43±1.63	83.12±0.48	82.56±1.47	79.17±2.77	73.33±1.15	82.01±1.81	83.25±1.61	**84.17±1.65**
Kappa(%)	81.76±1.14	77.84±1.11	80.22±0.51	75.04±1.02	80.11±0.53	80.31±0.55	79.71±0.72	76.75±0.71	69.62±1.17	80.61±0.19	80.83±0.15	**83.51±0.65**

dataset using the SVM, KNN, and RF classifiers. Fig. 5(d)–(f) shows the curves of the OA obtained by different band subsets and AllBands through three classifiers.

From Fig. 5(d), it can be seen that the DISGT initially reveals a similar OA to BS-Net, DARecNet, and ASPS at the beginning. However, when b is greater than 15, the quality of the bands selected by the DISGT surpasses that of other methods. It is worth noting that the OA of the DISGT outperforms AllBands when b is only 19, while other methods either do not compete with AllBands or require more bands to approximate its performance. This indicates that the DISGT more efficiently eliminates redundant bands. Fig. 5(e) and (f)

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TABLE IV
CLASSIFICATION ACCURACY OF DIFFERENT BS METHODS USING 23 BANDS ON THE PAVIA UNIVERSITY DATASET. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

No.	AllBands	ASPS	FNGBS	GRSC	BS-Net	DARecNet	BHCNN	BSD-GCN	MGSR	MR-Selection	S^4P	DISGT
1	94.12±0.41	91.77±0.84	91.80±1.17	92.64±0.61	91.74±0.57	91.91±0.15	90.97±0.81	90.32±0.87	73.21±0.18	90.78±0.43	93.47±0.47	**94.31±0.32**
2	96.71±0.38	94.11±0.47	95.25±0.21	94.88±0.24	95.41±0.43	91.51±0.28	94.81±1.37	90.46±0.31	76.18±0.01	96.11±0.39	96.31±0.04	**97.18±0.31**
3	84.44±1.52	76.81±1.57	76.81±0.35	79.14±1.46	79.81±1.18	78.41±1.47	79.45±0.51	73.96±0.53	46.01±1.38	76.26±3.16	**85.12±0.07**	84.91±0.28
4	97.27±0.61	95.18±1.51	94.68±0.57	93.71±0.67	96.51±0.65	93.31±0.96	92.72±0.38	90.11±1.37	97.03±0.51	**96.79±0.39**	73.64±0.83	95.97±1.91
5	99.75±0.32	**99.84±0.20**	99.71±0.15	99.61±0.36	99.71±0.23	99.55±0.57	98.71±0.53	99.34±0.43	99.87±0.17	99.55±0.29	97.65±0.87	99.81±0.15
6	91.33±1.24	83.83±0.61	85.65±1.77	86.14±1.01	86.35±2.27	84.31±0.92	85.21±0.58	74.34±0.87	68.94±0.92	92.40±1.28	91.78±0.39	**92.41±0.75**
7	86.68±3.24	82.84±1.88	81.65±1.14	84.73±2.11	79.68±0.82	82.21±1.48	84.11±0.43	81.45±2.5	56.84±1.77	77.11±3.61	83.96±1.52	**85.21±1.28**
8	86.11±1.08	85.21±0.83	84.15±1.28	84.37±0.91	86.35±0.72	84.64±1.33	85.41±0.56	83.54±1.25	65.21±1.35	83.38±0.42	86.72±0.31	**88.21±0.48**
9	99.99±0.05	99.92±0.18	99.77±0.24	99.63±0.13	99.74±0.28	99.81±0.13	99.64±0.31	99.48±1.11	81.43±1.42	99.89±0.15	99.89±0.11	**99.99±0.07**
OA(%)	93.82±0.14	92.30±0.24	92.55±0.13	90.77±0.14	92.12±0.13	92.51±0.15	91.31±0.33	87.47±0.32	73.25±0.21	92.49±0.29	93.82±0.15	**94.41±0.18**
AA(%)	92.84±0.72	89.75±0.12	89.97±0.25	90.61±0.23	90.68±0.35	88.25±0.28	90.24±0.13	86.95±0.71	70.93±0.51	90.32±0.77	92.62±0.35	**93.25±0.33**
Kappa(%)	92.15±0.41	87.98±0.21	88.73±0.15	88.91±0.23	89.42±0.18	84.81±0.13	88.55±0.48	83.34±0.43	62.61±0.26	90.02±0.38	91.78±0.16	**92.68±0.12**

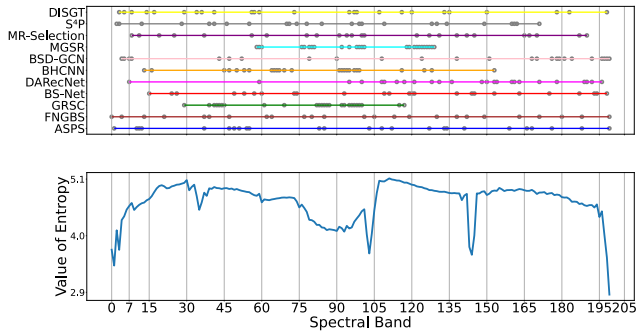


Fig. 7. Distribution of 25 bands selected by different BS methods on the Indian Pines dataset and the entropy value for each band.

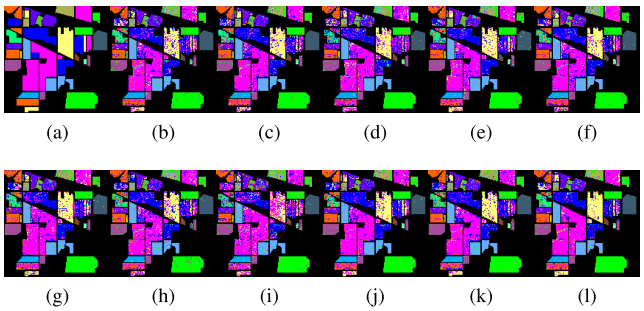


Fig. 8. Classification maps of different BS methods on the Indian Pines dataset using 25 bands with the SVM. (a) Ground truth. (b) ASPS. (c) FNGBS. (d) GRSC. (e) BS-Net. (f) DARecNet. (g) BHCNN. (h) BSD-GCN. (i) MGSR. (j) MR-Selection. (k) S^4P. (l) DISGT.

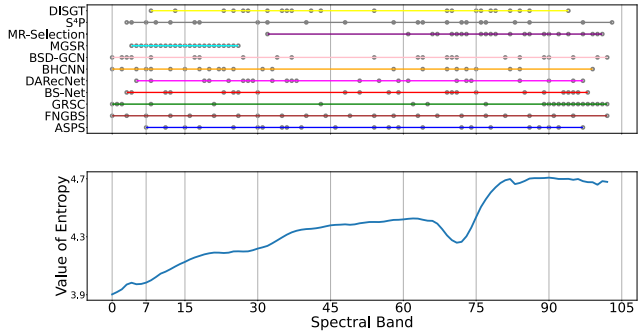


Fig. 9. Distribution of 23 bands selected by different BS methods on the Pavia University dataset and the entropy value for each band.

demonstrates a similar trend that the OA of the DISGT is almost the same as other methods when fewer than 17 bands are selected. However, the DISGT achieves results comparable to AllBands using fewer bands.

Table IV displays the classification accuracy of different methods when b is equal to 23, as well as AllBands. Among these methods, the DISGT exceeds the others in the OA, AA, and Kappa, achieving the highest classification accuracy for six classes. To better illustrate the differences between the band subsets of different methods, Fig. 9 visualizes their distribution. The classification maps of ten representative methods using 23 bands with the SVM are shown in Fig. 10. Overall, the band subset of the DISGT possesses a more even distribution and higher entropy values, resulting in a distinct classification map.

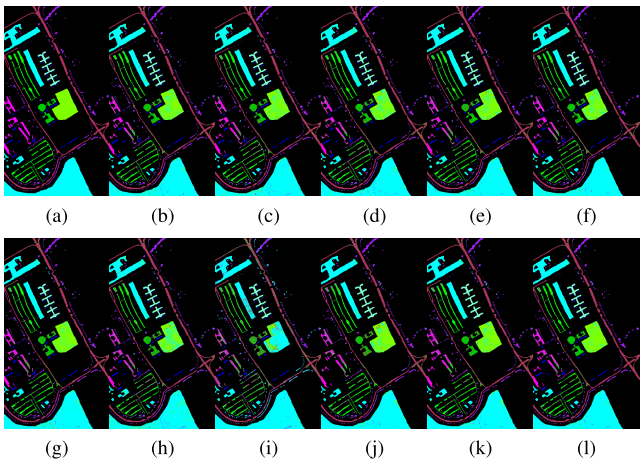


Fig. 10. Classification maps of different BS methods on the Pavia University dataset using 23 bands with SVM. (a) Ground truth. (b) ASPS. (c) FNGBS. (d) GRSC. (e) BS-Net. (f) DARecNet. (g) BHCNN. (h) BSD-GCN. (i) MGSR. (j) MR-Selection. (k) S^4P. (l) DISGT.

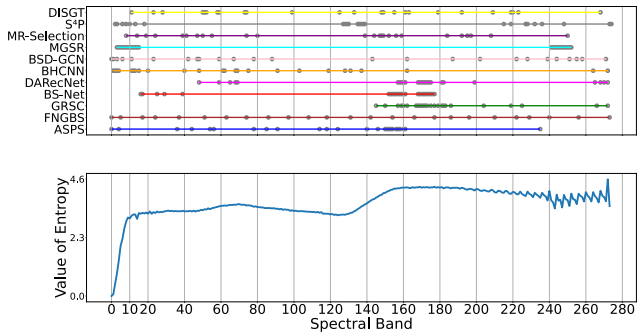


Fig. 11. Distribution of 25 bands selected by different BS methods on the Hanchuan dataset and the entropy value for each band.

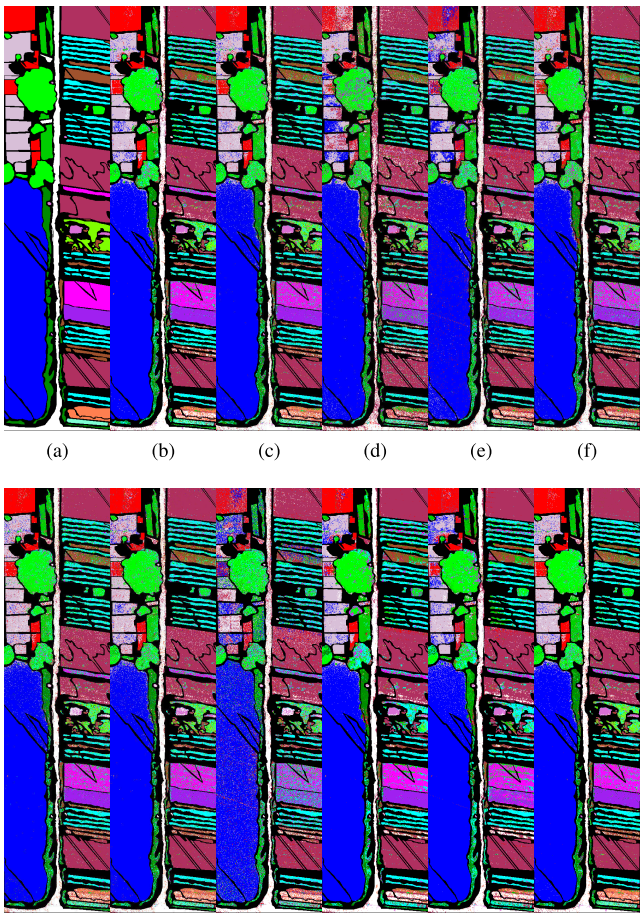
E. Results on the Hanchuan Dataset

The performance of the DISGT as well as other comparative methods and AllBands on the Hanchuan dataset is evaluated using three classifiers. The curves of the OA obtained with band subsets for different BS methods and AllBands on three different classifiers are shown in Fig. 5(g)–(i).

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TABLE V
CLASSIFICATION ACCURACY OF DIFFERENT BS METHODS USING 25 BANDS ON THE HANCHUAN DATASET. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

No.	AllBands	ASPS	FNGBS	GRSC	BS-Net	DARecNet	BHCNN	BSD-GCN	MGSR	MR-Selection	S^4P	DISGT
1	91.42±0.13	88.33±0.34	89.45±0.42	70.78±0.18	81.02±0.22	83.82±0.01	89.31±0.13	88.53±0.31	76.53±0.13	86.71±0.05	83.27±0.38	**90.24±0.14**
2	72.52±0.15	70.34±0.32	71.08±0.13	69.75±0.15	63.31±0.13	67.16±0.16	70.55±0.21	72.65±0.23	55.13±0.15	69.49±0.85	67.53±0.38	**74.53±0.33**
3	83.42±0.11	74.31±2.33	72.36±1.14	64.12±0.21	63.73±0.21	68.12±0.14	76.58±0.93	75.44±0.45	47.85±0.04	72.17±0.36	62.97±0.25	**78.51±0.55**
4	97.25±0.22	92.73±0.73	84.94±0.62	74.41±0.65	79.75±0.73	83.82±0.15	92.07±0.83	93.25±0.62	47.14±0.33	93.68±0.11	71.79±0.51	**93.83±0.54**
5	62.81±3.83	51.06±1.14	50.18±1.73	42.53±0.82	39.41±0.60	49.21±1.98	54.81±0.21	51.46±0.43	47.43±0.11	54.39±1.72	44.12±0.68	**60.15±0.03**
6	67.95±0.34	52.13±0.12	48.46±0.22	49.13±0.54	42.83±0.22	47.13±0.45	50.26±0.31	51.83±0.85	41.75±0.05	53.13±1.07	43.78±1.11	**63.37±0.11**
7	78.43±0.70	67.15±0.14	71.32±0.75	56.98±0.42	65.21±0.71	69.95±0.24	71.13±0.13	72.25±0.94	39.82±0.03	64.69±0.05	53.75±0.86	**76.51±0.35**
8	82.23±0.45	80.38±0.43	77.72±0.54	71.34±0.58	71.93±0.35	75.41±0.35	78.67±0.38	79.11±0.13	63.24±0.31	79.71±0.43	73.02±0.06	**82.63±0.15**
9	79.33±1.12	69.17±0.12	72.85±0.41	60.03±0.57	57.75±0.27	64.45±0.26	72.16±0.35	73.35±0.12	46.73±0.03	71.88±0.58	62.75±0.95	**75.84±0.23**
10	97.81±0.14	**94.03±0.14**	91.64±0.43	53.11±0.68	59.05±0.62	76.13±0.07	91.7±0.41	86.43±0.64	45.42±0.81	87.43±0.32	79.71±0.39	93.82±0.49
11	92.14±0.35	83.95±0.12	83.93±0.14	67.93±0.35	81.54±0.21	80.91±0.18	84.61±0.23	81.13±0.43	44.14±0.43	83.96±0.26	78.56±0.03	**86.47±0.14**
12	77.83±1.94	59.24±1.51	55.98±1.43	33.91±0.14	42.91±0.23	40.63±0.65	57.63±0.23	54.35±0.92	42.31±0.84	62.21±0.81	59.72±0.48	**73.21±0.13**
13	77.64±1.37	62.13±0.13	64.84±0.02	55.34±0.24	58.73±0.51	60.25±0.96	66.48±0.11	65.46±0.18	42.13±0.94	66.64±0.05	60.58±0.16	**74.93±0.54**
14	87.24±0.53	85.15±0.15	**85.35±0.15**	65.43±0.68	83.62±0.13	81.67±0.25	82.5±1.45	84.11±0.34	72.63±0.28	83.49±0.07	81.19±0.21	85.24±0.38
15	97.41±1.51	80.64±2.47	84.43±1.63	67.55±0.35	76.48±1.54	76.53±2.75	85.38±0.03	82.23±1.23	89.81±0.97	86.32±2.09	82.08±1.83	**92.13±1.74**
16	99.52±0.05	98.12±0.15	98.05±0.53	95.28±0.13	92.79±0.13	98.13±0.21	98.41±0.13	97.88±0.17	87.63±0.13	97.52±0.01	96.22±0.09	**98.57±0.13**
OA(%)	89.36±0.13	85.34±0.43	84.95±0.15	74.84±0.04	77.87±0.13	81.42±0.13	85.35±0.13	84.85±0.16	67.38±0.15	84.63±0.06	80.46±0.13	**88.92±0.11**
AA(%)	83.54±0.12	75.53±0.12	75.17±0.12	62.32±0.05	66.29±0.14	70.21±0.44	76.33±0.12	75.53±0.15	55.65±0.03	75.84±0.11	68.82±0.12	**81.25±0.65**
Kappa(%)	87.31±0.27	83.35±0.95	82.35±0.03	70.31±0.13	73.85±0.03	78.23±0.15	82.83±0.14	82.21±0.13	61.73±0.11	81.96±0.08	77.02±0.15	**84.71±0.13**



(a) (b) (c) (d) (e) (f)

(g) (h) (i) (j) (k) (l)

Fig. 12. Classification maps of different BS methods on the Hanchuan dataset using 25 bands with SVM. (a) Ground truth. (b) ASPS. (c) FNGBS. (d) GRSC. (e) BS-Net. (f) DARecNet. (g) BHCNN. (h) BSD-GCN. (i) MGSR. (j) MR-Selection. (k) S^4P. (l) DISGT.

As shown in Fig. 5(g), when b is less than 17, the selected bands of the DISGT are close to FNGBS, BHCNN, and MR-Selection. As b exceeds 19, the OA of the DISGT consistently outperforms others without showing a significant decreasing trend. Similarly, in Fig. 5(h) and (i), the DISGT demonstrates preferable results compared to other methods. It is worth noting that the Hanchuan dataset, with its 274 bands and 1217 × 303 pixels per band, carries more meaningful information than other datasets. Consequently, the classification performance when selecting fewer bands is inferior to that of AllBands.

TABLE VI
ABLATION ANALYSIS OF THREE DATASETS WITH THE mOA (%)

Method	Indian Pines	Pavia University	Hanchuan
DISGT	**83.23**	**92.18**	**82.16**
DISGT-NS	77.39	90.05	81.57
DISGT-NI	81.25	90.49	80.46

The classification accuracy, including AllBands with b set to 25, on the Hanchuan dataset is presented in Table V. The proposed DISGT demonstrates better performance for most classes. In addition, the distribution of 25 bands selected by different BS methods and the entropy value for each band are shown in Fig. 11. The classification maps for different methods using 25 bands with the SVM are displayed in Fig. 12. As observed, the DISGT selects more crucial bands in a decentralized manner, resulting in better performance in BS when processing high-dimensional datasets with a large number of samples.

F. Ablation Experiments

The proposed DISGT utilizes the idea of dynamic programming to reduce time and space complexities. Two variants of the DISGT are created to investigate how the performance is improved for our method. DISGT-NS is a method without the DSP module and constructs the entire HSI into a graph. DISGT-NI is a method without the SII module. Table VI reveals the mOA of the DISGT as well as its two variants using the SVM classifier. Bands ranging from 5 to 30 were selected to calculate the mOA for each method. The results are presented in Table VI, with the optimal mOA values highlighted in bold. The addition of the DSP and SII modules significantly enhances the BS ability. This is because the DSP module allocates the homogeneous bands into subspaces by using the dynamic distance norm, which effectively distinguishes the regional differences of spectra. The SII module shares bands with uncertain subspace membership to other subspaces and considers global information by establishing relationships between local information in subspaces. Constructing subgraphs from these interacted subspaces accurately

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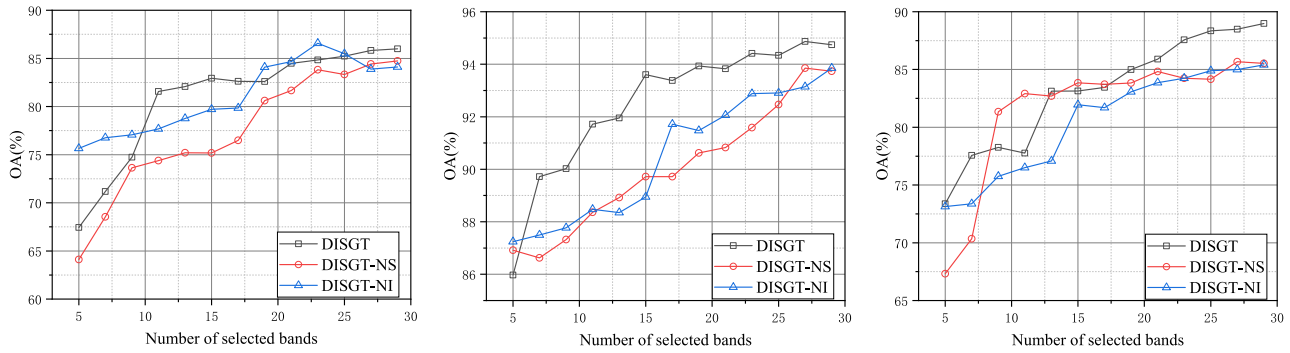


Fig. 13. SVM-based classification performance of the DISGT and its variants of OA under a different number of selected bands. (a) Indian Pines. (b) Pavia University. (c) Hanchuan.

TABLE VII
COMPUTATION TIME (S) FOR SELECTING 19 BANDS USING DIFFERENT METHODS

Methods	Indian Pines	Pavia University	Hanchuan
ASPS	0.31	0.66	2.11
FNGBS	1.51	7.57	493.66
GRSC	7.69	48.42	673.07
MGSR	18.02	165.51	702.97
S^4P	3.21	8.73	690.39
BS-Net	372.07	810.38	>3h
DARecNet	321.57	762.51	>3h
BHCNN	603.1	2078.91	>3h
BSD-GCN	3461.15	>1h	>3h
MR-Selection	238.16	578.37	7986.11
DISGT	201.67	501.27	5489.20

reflects the spectral structure of HSI, thereby fully utilizing the differences in the distributions of bands within and between subgraphs. Without the DSP module, directly processing high-dimensional data requires more computational resources and time. In addition, considering the spectrum as a whole inevitably overlooks important underlying features, affecting classification accuracy. On the other hand, including DSP without the SII model means partitioning subspaces without considering the relationships between them, which is detrimental to capturing the global spectral information, resulting in insufficient feature expression and decreased classification accuracy.

To further evaluate the effectiveness of the proposed method, the performance of the DISGT and its two variants are tested on three datasets. As shown in Fig. 13, in most cases, the DISGT obtains a higher OA than the other two variants. This demonstrates the necessity of incorporating both DSP and SII modules.

G. Analysis of Running Time

To analyze the running time, all comparative methods are executed in the same environment, and their running times are recorded for selecting 19 bands across three datasets.

The ASPS, FNGBS, GRSC, MGSR, and S^4P are executed in MATLAB and other methods are implemented in Python via GPU. The final results are shown in Table VII. It is noticed that the DISGT shows an obvious time advantage over other DNN-based methods, especially when dealing with large-sample datasets like Hanchuan. The subgraphs are applied to divide all bands into several subgraphs for quick processing, whereas other DNN-based methods require more time. In contrast to ASPS, FNGBS, GRSC, MGSR, and S^4P, the latter six methods exhibit greater computational complexity because they require more time to train the model. However, it is important to note that the DISGT acquired the best classification accuracy under the same conditions. This suggests that the former four methods incompletely search band subsets, whereas DISGT provides a more thorough and effective approach.

In conclusion, the DISGT strikes a balance between classification accuracy and running time, realizing a high-quality BS task within an acceptable time and space complexity.

V. CONCLUSION

This article proposes an unsupervised BS method called DISGT, which adaptively obtains the optimal band subset while reducing the time and space complexities of BS. To achieve this, the DISGT considers the differences of heterogeneous bands from the spectral perspective and dynamically allocates them to several subspaces. Moreover, the relationships between different subspaces are established by information interaction to enhance the diversity of band combinations. These subspaces are constructed into subgraphs to improve the correlation between distant and neighboring bands. The SGT is used to handle subgraphs with different numbers of bands, efficiently extracting both local information from subspaces and global information from spectra.

In summary, the proposed DISGT provides a novel and effective unsupervised BS method, demonstrating superior performance and robustness compared to other BS methods on the Indian Pines, Pavia University, and Hanchuan datasets. Future work will focus on directly obtaining the optimal number of subspaces for a new HSI without requiring independent experiments. In addition, as with other BS methods, how to determine the optimal number of band subsets is a major challenge. Thus, we will explore further solutions to these problems in the future.

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